

Dissecting the cotranscriptome landscape of plants and their microbiota

Tatsuya Nobori^{4,5}, Yu Cao⁴ , Frederickson Entila⁴ , Eik Dahms⁴ , Yayoi Tsuda^{1,2,3,4},
Ruben Garrido-Oter^{4,6} & Kenichi Tsuda^{1,2,3,4,*} 

Abstract

Interactions between plants and neighboring microbial species are fundamental elements that collectively determine the structure and function of the plant microbiota. However, the molecular basis of such interactions is poorly characterized. Here, we colonize *Arabidopsis* leaves with nine plant-associated bacteria from all major phyla of the plant microbiota and profile cotranscriptomes of plants and bacteria six hours after inoculation. We detect both common and distinct cotranscriptome signatures among plant–commensal pairs. *In planta* responses of commensals are similar to those of a disarmed pathogen characterized by the suppression of genes involved in general metabolism in contrast to a virulent pathogen. We identify genes that are enriched in the genome of plant-associated bacteria and induced *in planta*, which may be instrumental for bacterial adaptation to the host environment and niche separation. This study provides insights into how plants discriminate among bacterial strains and lays the foundation for in-depth mechanistic dissection of plant–microbiota interactions.

Keywords *Arabidopsis*; bacterial adaptation; commensal; cotranscriptome; plant microbiota

Subject Categories Chromatin, Transcription, & Genomics; Microbiology, Virology & Host Pathogen Interaction; Plant Biology

DOI 10.15252/embr.202255380 | Received 10 May 2022 | Revised 9 September 2022 | Accepted 13 September 2022 | Published online 11 October 2022

EMBO Reports (2022) 23: e55380

Introduction

In nature, plants assemble bacterial communities with well-defined taxonomic structures (the plant microbiota) (Hacquard *et al.*, 2015), which can be harnessed for plant health and survival (Durán *et al.*, 2018; Kwak *et al.*, 2018; Carrión *et al.*, 2019). How plants discriminate among various bacterial strains and establish strain-

specific associations in a community context remains an open question in basic plant microbiota research and is key to facilitating the application of microbiota-based strategies to improve plant health in agricultural settings. Answering this question requires a comprehensive and unified understanding of plant and bacterial responses during their interactions.

The plant innate immune system contributes to the assembly and maintenance of healthy bacterial communities (Lebeis *et al.*, 2015; Chen *et al.*, 2020). A crucial part of the plant immune system is the perception of environmental microbes using cell surface receptors that detect conserved microbial epitopes, termed microbe-associated molecular patterns (MAMPs) (Boller & Felix, 2009). Recognition of MAMPs triggers defense responses collectively called pattern-triggered immunity (PTI), which can inhibit pathogen growth (Jones & Dangl, 2006). MAMPs such as the bacterial flagellin peptide flg22 are widely conserved in nonpathogenic bacteria and pathogenic bacteria (Hacquard *et al.*, 2017), and some nonpathogenic Proteobacteria strains were shown to trigger defense responses in plant leaves likely via PTI pathways (Vogel *et al.*, 2016). On the contrary, diverse microbiota members can suppress PTI triggered by flg22 in roots (Garrido-Oter *et al.*, 2018; Yu *et al.*, 2019; Teixeira *et al.*, 2021), which can facilitate colonization by the root microbiota (Yu *et al.*, 2019; Teixeira *et al.*, 2021). Thus, PTI activation by divergent MAMPs and subsequent PTI modulation by plant-associated bacteria might steer plant responses in a bacterial strain-specific manner, contributing to microbiota assembly in plants. A recent study identified, in the aboveground part of plants, a core set of genes induced by phylogenetically diverse endogenous bacteria; some of these genes contribute to plant defense against pathogens (Maier *et al.*, 2021). Therefore, studying common and specific plant responses to diverse bacteria is crucial to understand the role of the plant immune system in the face of both pathogenic and non-pathogenic microbes.

When colonized densely and heterogeneously by various bacterial species, plants might not be able to tailor their responses to individual bacterial strains. Yet, it might be possible that the same

1 State Key Laboratory of Agricultural Microbiology, Hubei Hongshan Laboratory, Hubei Key Lab of Plant Pathology, College of Plant Science and Technology, Huazhong Agricultural University, Wuhan, China

2 Shenzhen Institute of Nutrition and Health, Huazhong Agricultural University, Wuhan, China

3 Shenzhen Branch, Guangdong Laboratory of Lingnan Modern Agriculture, Genome Analysis Laboratory of the Ministry of Agriculture and Rural Affairs, Agricultural Genomics Institute at Shenzhen, Chinese Academy of Agricultural Sciences, Shenzhen, China

4 Department of Plant Microbe Interactions, Max Planck Institute for Plant Breeding Research, Cologne, Germany

5 Salk Institute for Biological Studies, La Jolla, CA, USA

6 Cluster of Excellence on Plant Sciences, Düsseldorf, Germany

*Corresponding author. E-mail: tsuda@mail.hzau.edu.cn